

Simple Python Programs

1. Given two integer variables `a` and `b`, assign their sum to a variable `c`.
2. Given two variables `x` and `y`, swap their values using a temporary variable `temp`.
3. Given two integer variables `x` and `y`, swap their values without using a third variable.
4. Calculate the area of a rectangle given variables `length` and `width`.
5. Calculate the perimeter of a rectangle given variables `length` and `width`.
6. Given a variable `radius`, calculate the circumference of a circle (use 3.14 for pi).
7. Given a variable `radius`, calculate the area of a circle.
8. Given `base` and `height`, calculate the area of a triangle.
9. Given a variable `n`, extract and store its last digit in a variable `last\_digit`.
10. Given a variable `n`, remove its last digit and store the new number in `n`.
11. Given three variables `m1`, `m2`, and `m3` representing test marks, calculate their average.
12. Given a principal amount `P`, rate `R`, and time `T`, calculate the simple interest.
13. Given a variable `celsius`, convert it to Fahrenheit and store it in `fahrenheit`.
14. Given a variable `fahrenheit`, convert it to Celsius and store it in `celsius`.
15. Given a total number of days in a variable `days`, calculate the number of weeks and assign it to `weeks`.
16. Given a total number of days, find the remaining days after extracting full weeks.
17. Given a number of seconds, calculate how many full minutes it contains.
18. Given a number of seconds, find the remaining seconds after extracting full minutes.
19. Given three integer variables `a`, `b`, and `c`, calculate the sum of their squares.
20. Given a total price and a discount percentage, calculate the final price after the discount.
21. Given the basic salary of an employee, calculate HRA as 20% of basic and DA as 50% of basic, and store the gross salary.
22. Given a variable `num`, extract the second-to-last digit of the number.
23. Given a two-digit integer, reverse its digits mathematically.
24. Given a three-digit integer, calculate the sum of its digits.
25. Given a number, multiply it by 2 using a bitwise left shift operator.
26. Given a number, divide it by 2 using a bitwise right shift operator.
27. Given a number, find its remainder when divided by 8 using bitwise AND.
28. Given two numbers, find their bitwise XOR and store the result.
29. Given a number, toggle its lowest bit using bitwise XOR.
30. Given two variables `a` and `b`, compute the result of $(a + b)^2$ mathematically.
31. Given the Cartesian coordinates `(x1, y1)` and `(x2, y2)`, calculate the square of the distance between them.
32. Given a weight in kilograms, convert it to grams.
33. Given a weight in pounds, convert it to kilograms (use 1 lb = 0.453592 kg).
34. Calculate the volume of a cylinder given its `radius` and `height`.
35. Calculate the surface area of a cube given the length of its `side`.
36. Given a number of bytes, convert it to kilobytes.
37. Given a boolean variable `is\_raining`, assign its logical NOT to `go\_outside`.

38. Given two boolean variables ``has_ticket`` and ``is_vip``, assign the logical OR of these to ``can_enter``.
39. Given two boolean variables ``is_logged_in`` and ``has_permission``, assign their logical AND to ``show_dashboard``.
40. Check if a number ``x`` is greater than 10 and less than 20, storing the boolean result in ``is_valid``.
41. Check if a given year is divisible by 4, storing the boolean result in ``is_leap_candidate``.
42. Given an integer, store ``True`` in a variable if it is even, using the modulo operator.
43. Given an integer, store ``True`` in a variable if it is odd, using bitwise AND.
44. Given a variable ``x``, apply the pre-increment operator and assign the result to ``y``.
45. Given a variable ``a``, apply the post-increment operator and assign the result to ``b``.
46. Given a variable ``score``, increase its value by 10 using a compound assignment operator.
47. Given a variable ``balance``, decrease its value by 50 using a compound assignment operator.
48. Given an initial velocity ``u``, acceleration ``a``, and time ``t``, calculate final velocity using $v = u + at$.
49. Given initial velocity ``u``, time ``t``, and acceleration ``a``, calculate distance using $s = ut + 0.5at^2$.
50. Given mass ``m`` and velocity ``v``, calculate kinetic energy.
51. Calculate the Body Mass Index (BMI) given ``weight`` in kg and ``height`` in meters.
52. Given the lengths of two parallel sides ``a`` and ``b``, and height ``h``, calculate the area of a trapezoid.
53. Given a number ``n``, calculate its cube without using built-in power functions.
54. Given a number of hours, minutes, and seconds, calculate the total equivalent seconds.
55. Given a 4-digit number, calculate the product of its first and last digit.
56. Given an amount in cents, determine how many full quarters (25 cents) can be made.
57. Given an amount in cents, determine the remaining cents after extracting full quarters.
58. Given a variable ``n``, clear its lowest set bit using bitwise operations.
59. Given two numbers ``x`` and ``y``, store ``True`` if they are strictly equal, otherwise ``False``.
60. Given two numbers ``x`` and ``y``, store ``True`` if ``x`` is not equal to ``y``.
61. Given a variable ``age``, store ``True`` if it is greater than or equal to 18.
62. Given variables ``cost_price`` and ``selling_price``, calculate the absolute profit amount.
63. Given ``cost_price`` and ``selling_price``, store ``True`` if there is a profit, otherwise ``False``.
64. Given a variable ``angle``, convert it from degrees to radians (using 3.14 for pi).
65. Given a variable ``radians``, convert it from radians to degrees.
66. Given a total number of items and a box capacity, calculate how many full boxes are completely filled.
67. Given a total number of items and a box capacity, calculate the number of items left over that don't fill a box.
68. Calculate the volume of a sphere given its ``radius``.
69. Given the sides of a triangle ``a``, ``b``, and ``c``, calculate its semi-perimeter ``s``.
70. Given the semi-perimeter ``s`` and sides ``a``, ``b``, ``c``, calculate the squared area of the triangle using Heron's formula (without applying the square root).
71. Given two numbers ``x`` and ``y``, determine the result of $(x \& y) | (x \wedge y)$.

72. Given a variable ``n``, multiply it by 8 using only bitwise operators.
73. Given a number ``N``, isolate its lowest set bit using ``N & -N``.
74. Given three sides of a cuboid ``l``, ``w``, and ``h``, calculate its total surface area.
75. Given three variables ``x``, ``y``, and ``z``, check if ``x`` is the largest using only logical and relational operators, storing the boolean result.
76. Given a lowercase character's ASCII integer value, subtract 32 to convert it to an uppercase ASCII value.
77. Given an uppercase character's ASCII integer value, add 32 to convert it to a lowercase ASCII value.
78. Given two fractional parts ``num1`/`den1`` and ``num2`/`den2``, calculate the numerator of their sum.
79. Given two fractional parts ``num1`/`den1`` and ``num2`/`den2``, calculate the denominator of their sum.
80. Given a total number of months, calculate the equivalent number of full years and remaining months.
81. Given a bill amount and a tip percentage, calculate the total amount to be paid.
82. Given a bill amount, tip percentage, and number of people, calculate the equally split cost per person.
83. Calculate the length of the hypotenuse squared given the two other sides of a right-angled triangle ``a`` and ``b``.
84. Given ``N`` items distributed equally among ``K`` people, calculate how many items each person receives.
85. Given ``N`` items distributed equally among ``K`` people, calculate how many items remain undistributed.
86. Given the coordinates of a circle's center ``(cx, cy)`` and a point ``(px, py)`` on the circumference, calculate the squared radius.
87. Given an integer ``num``, use bitwise NOT to find its 1's complement.
88. Given an integer ``num``, calculate its 2's complement using bitwise NOT and addition.
89. Given a variable ``val``, store ``True`` if it is a multiple of both 3 and 5.
90. Given a total string character length and a character limit per line, calculate the maximum number of full lines.
91. Given the internal and external radii of a hollow cylinder and its height, calculate the volume of material used.
92. Given a three-digit number, calculate the number formed by its digits in reverse order mathematically.
93. Check if a point ``(x, y)`` lies in the first quadrant, storing the boolean result in a variable.
94. Check if a point ``(x, y)`` lies exactly at the origin, storing the boolean result in a variable.
95. Given a file size in megabytes, convert it to bits.
96. Given the circumference of a circle, calculate its diameter.
97. Given an employee's hourly wage and hours worked, calculate their total weekly pay.
98. Given a car's fuel efficiency (miles per gallon) and fuel tank capacity, calculate its maximum range in miles.
99. Given a person's birth year and the current year, calculate their approximate age.

100. Given two integers `a` and `b`, check if their sum is exactly zero and store the boolean result.